

# Cálculo Multivariável

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## Seção 14.4 — Regra da Cadeia

Exercícios: 2, 3, 4, 6, 7, 8, 9, 13, 14, 15

Resoluções completas com justificativas

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## Exercícios 1–6: Expresse $dw/dt$ como função de $t$ ; calcule $dw/dt$ no ponto dado

**Método:** Pela regra da cadeia,  $\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt}$  (ou com mais variáveis). Após derivar, substitui-se  $x, y, z$  em termos de  $t$ .

**Exercício 2**  $w = x^2 + y^2$ ,  $x = \cos t + \sin t$ ,  $y = \cos t - \sin t$ ;  $t = 0$

(a) Pela regra da cadeia:

$$\frac{dw}{dt} = 2x(-\sin t + \cos t) + 2y(-\sin t - \cos t)$$

Substituindo  $x$  e  $y$ :

$$\begin{aligned} &= 2(\cos t + \sin t)(\cos t - \sin t) + 2(\cos t - \sin t)(-\sin t - \cos t) \\ &= 2(\cos^2 t - \sin^2 t) - 2(\cos^2 t - \sin^2 t) = 0 \end{aligned}$$

**Verificação direta:**  $w = (\cos t + \sin t)^2 + (\cos t - \sin t)^2 = 2\cos^2 t + 2\sin^2 t = 2$  (constante), logo  $dw/dt = 0$ .

(b)  $\left. \frac{dw}{dt} \right|_{t=0} = \boxed{0}$

**Exercício 3**  $w = \frac{x}{z} + \frac{y}{z}$ ,  $x = \cos^2 t$ ,  $y = \sin^2 t$ ,  $z = \frac{1}{t}$ ;  $t = 3$

(a) Derivadas parciais:  $\frac{\partial w}{\partial x} = \frac{1}{z}$ ,  $\frac{\partial w}{\partial y} = \frac{1}{z}$ ,  $\frac{\partial w}{\partial z} = -\frac{x+y}{z^2}$ .

Derivadas em relação a  $t$  (usando  $\sin 2t = 2\sin t \cos t$ ):

$$\frac{dx}{dt} = -\sin 2t, \quad \frac{dy}{dt} = \sin 2t, \quad \frac{dz}{dt} = -\frac{1}{t^2}$$

Pela regra da cadeia:

$$\frac{dw}{dt} = \frac{1}{z}(-\sin 2t) + \frac{1}{z}(\sin 2t) + \left(-\frac{x+y}{z^2}\right)\left(-\frac{1}{t^2}\right)$$

Os dois primeiros termos se cancelam. Usando  $x + y = \cos^2 t + \sin^2 t = 1$  e  $z^2 = 1/t^2$ :

$$= \frac{x+y}{z^2 t^2} = \frac{1}{(1/t^2) \cdot t^2} = 1$$

**Verificação direta:**  $w = \frac{x+y}{z} = \frac{1}{1/t} = t$ , portanto  $dw/dt = 1$  trivialmente.

$$(b) \left. \frac{dw}{dt} \right|_{t=3} = \boxed{1}$$

**Exercício 4**  $w = \ln(x^2 + y^2 + z^2)$ ,  $x = \cos t$ ,  $y = \sin t$ ,  $z = 4\sqrt{t}$ ;  $t = 3$

(a) Derivadas parciais:  $\frac{\partial w}{\partial x} = \frac{2x}{x^2 + y^2 + z^2}$  (e análogos para  $y, z$ ).

Derivadas em relação a  $t$ :

$$\frac{dx}{dt} = -\sin t, \quad \frac{dy}{dt} = \cos t, \quad \frac{dz}{dt} = \frac{2}{\sqrt{t}}$$

Pela regra da cadeia — numerador:

$$2x(-\sin t) + 2y(\cos t) + 2z \cdot \frac{2}{\sqrt{t}}$$

Substituindo  $x = \cos t$ ,  $y = \sin t$ :

$$= \underbrace{-2 \cos t \sin t + 2 \sin t \cos t}_{=0} + 2 \cdot 4\sqrt{t} \cdot \frac{2}{\sqrt{t}} = 16$$

Denominador:  $x^2 + y^2 + z^2 = \underbrace{\cos^2 t + \sin^2 t}_{=1} + 16t = 1 + 16t$ .

$$\boxed{\frac{dw}{dt} = \frac{16}{1 + 16t}}$$

$$(b) \left. \frac{dw}{dt} \right|_{t=3} = \frac{16}{1 + 48} = \boxed{\frac{16}{49}}$$

**Exercício 6**  $w = z - \sin xy$ ,  $x = t$ ,  $y = \ln t$ ,  $z = e^{t-1}$ ;  $t = 1$

(a) Derivadas parciais:  $\frac{\partial w}{\partial x} = -y \cos xy$ ,  $\frac{\partial w}{\partial y} = -x \cos xy$ ,  $\frac{\partial w}{\partial z} = 1$ .

Derivadas em relação a  $t$ :  $\frac{dx}{dt} = 1$ ,  $\frac{dy}{dt} = \frac{1}{t}$ ,  $\frac{dz}{dt} = e^{t-1}$ .

Pela regra da cadeia, substituindo  $x = t$ ,  $y = \ln t$ :

$$\frac{dw}{dt} = -\ln t \cos(t \ln t) - \frac{t}{t} \cos(t \ln t) + e^{t-1} = -(1 + \ln t) \cos(t \ln t) + e^{t-1}$$

(b) Em  $t = 1$ :  $x = 1$ ,  $y = \ln 1 = 0$ ,  $z = e^0 = 1$ :

$$= -(1 + 0) \cos(0) + e^0 = -1 + 1 = \boxed{0}$$

### Exercícios 7–8: Expresse $\partial z/\partial u$ e $\partial z/\partial v$ ; calcule no ponto dado

**Exercício 7**  $z = 4e^x \ln y$ ,  $x = \ln(u \cos v)$ ,  $y = u \sin v$ ;  $(u, v) = (2, \pi/4)$

(a) Derivadas parciais de  $z$ :

$$\frac{\partial z}{\partial x} = 4e^x \ln y, \quad \frac{\partial z}{\partial y} = \frac{4e^x}{y}$$

Derivadas de  $x$  e  $y$ :

$$\frac{\partial x}{\partial u} = \frac{1}{u}, \quad \frac{\partial x}{\partial v} = -\tan v, \quad \frac{\partial y}{\partial u} = \sin v, \quad \frac{\partial y}{\partial v} = u \cos v$$

Usando  $e^x = e^{\ln(u \cos v)} = u \cos v$  e  $y = u \sin v$ :

$$\frac{\partial z}{\partial u} = 4u \cos v \ln(u \sin v) \cdot \frac{1}{u} + \frac{4u \cos v}{u \sin v} \cdot \sin v = 4 \cos v [\ln(u \sin v) + 1]$$

$$\frac{\partial z}{\partial v} = -4u \sin v \ln(u \sin v) + \frac{4u \cos^2 v}{\sin v}$$

(b) Em  $(u, v) = (2, \pi/4)$ :  $\cos(\pi/4) = \sin(\pi/4) = \frac{\sqrt{2}}{2}$ ,  $u \sin v = \sqrt{2}$ ,  $\ln \sqrt{2} = \frac{\ln 2}{2}$ :

$$\frac{\partial z}{\partial u} = 4 \cdot \frac{\sqrt{2}}{2} \left( \frac{\ln 2}{2} + 1 \right) = \boxed{\sqrt{2} \ln 2 + 2\sqrt{2}}$$

$$\frac{\partial z}{\partial v} = -2\sqrt{2} \ln 2 + \frac{8}{\sqrt{2}} = \boxed{\sqrt{2}(4 - 2 \ln 2)}$$

**Exercício 8**  $z = \tan^{-1}(x/y)$ ,  $x = u \cos v$ ,  $y = u \sin v$ ;  $(u, v) = (1, 3, \pi/6)$

(a) Derivadas parciais de  $z$  (usando  $\frac{d}{du} \tan^{-1}(u) = \frac{1}{1+u^2}$ , simplificando  $\frac{1}{1+(x/y)^2} = \frac{y^2}{x^2+y^2}$ ):

$$\frac{\partial z}{\partial x} = \frac{y}{x^2 + y^2}, \quad \frac{\partial z}{\partial y} = -\frac{x}{x^2 + y^2}$$

Derivadas de  $x$  e  $y$ :

$$\frac{\partial x}{\partial u} = \cos v, \quad \frac{\partial x}{\partial v} = -u \sin v, \quad \frac{\partial y}{\partial u} = \sin v, \quad \frac{\partial y}{\partial v} = u \cos v$$

**Simplificação chave:**  $x^2 + y^2 = u^2 \cos^2 v + u^2 \sin^2 v = u^2 \underbrace{(\cos^2 v + \sin^2 v)}_{=1} = u^2$ .

Para  $\partial z / \partial u$ , substituindo  $x = u \cos v$ ,  $y = u \sin v$  e  $x^2 + y^2 = u^2$ :

$$\frac{\partial z}{\partial u} = \frac{u \sin v \cos v - u \cos v \sin v}{u^2} = \frac{0}{u^2} = \boxed{0}$$

Para  $\partial z / \partial v$ :

$$\frac{\partial z}{\partial v} = \frac{-u \sin v \cdot u \sin v - u \cos v \cdot u \cos v}{u^2} = \frac{-u^2(\sin^2 v + \cos^2 v)}{u^2} = \boxed{-1}$$

(b) Como os resultados são constantes, independem do ponto:

$$\left. \frac{\partial z}{\partial u} \right|_{(1, 3, \pi/6)} = \boxed{0} \quad \left. \frac{\partial z}{\partial v} \right|_{(1, 3, \pi/6)} = \boxed{-1}$$

**Interpretação:**  $z = \tan^{-1}\left(\frac{u \cos v}{u \sin v}\right) = \tan^{-1}(\cot v) = \frac{\pi}{2} - v$ , que depende apenas de  $v$  linearmente — por isso  $\partial z/\partial u = 0$  e  $\partial z/\partial v = -1$  são constantes.

### Exercícios 9–10: Expresse $\partial w/\partial u$ e $\partial w/\partial v$ ; calcule no ponto dado

**Exercício 9**  $w = xy + yz + xz$ ,  $x = u + v$ ,  $y = u - v$ ,  $z = uv$ ;  $(u, v) = (1/2, 1)$

(a) Derivadas parciais de  $w$ :

$$\frac{\partial w}{\partial x} = y + z, \quad \frac{\partial w}{\partial y} = x + z, \quad \frac{\partial w}{\partial z} = x + y$$

Derivadas de  $x, y, z$ :

$$\frac{\partial x}{\partial u} = 1, \quad \frac{\partial x}{\partial v} = 1, \quad \frac{\partial y}{\partial u} = 1, \quad \frac{\partial y}{\partial v} = -1, \quad \frac{\partial z}{\partial u} = v, \quad \frac{\partial z}{\partial v} = u$$

Substituindo  $x = u + v$ ,  $y = u - v$ ,  $z = uv$ :

$$\frac{\partial w}{\partial u} = (u - v + uv) + (u + v + uv) + (2u) \cdot v = 2u + 4uv$$

$$\frac{\partial w}{\partial v} = (u - v + uv) - (u + v + uv) + (2u) \cdot u = 2u^2 - 2v$$

**Verificação direta:**  $w = (u + v)(u - v) + (u - v)(uv) + (u + v)(uv) = u^2 - v^2 + 2u^2v$ , daí  $\partial w/\partial u = 2u + 4uv$  e  $\partial w/\partial v = 2u^2 - 2v$ . ✓

(b) Em  $(u, v) = (1/2, 1)$ :

$$\frac{\partial w}{\partial u} = 2 \cdot \frac{1}{2} + 4 \cdot \frac{1}{2} \cdot 1 = 1 + 2 = \boxed{3}$$

$$\frac{\partial w}{\partial v} = 2 \cdot \frac{1}{4} - 2 \cdot 1 = \frac{1}{2} - 2 = \boxed{-\frac{3}{2}}$$

**Exercício 10**  $w = \ln(x^2 + y^2 + z^2)$ ,  $x = ue^v \sin u$ ,  $y = ue^v \cos u$ ,  $z = ue^v$ ;  $(u, v) = (-2, 0)$

(a) Simplificação de  $x^2 + y^2 + z^2$ :

$$= u^2 e^{2v} \sin^2 u + u^2 e^{2v} \cos^2 u + u^2 e^{2v} = u^2 e^{2v} + u^2 e^{2v} = 2u^2 e^{2v}$$

Portanto:

$$w = \ln(2u^2 e^{2v}) = \ln 2 + 2 \ln |u| + 2v$$

$$\boxed{\frac{\partial w}{\partial u} = \frac{2}{u}}$$

$$\boxed{\frac{\partial w}{\partial v} = 2}$$

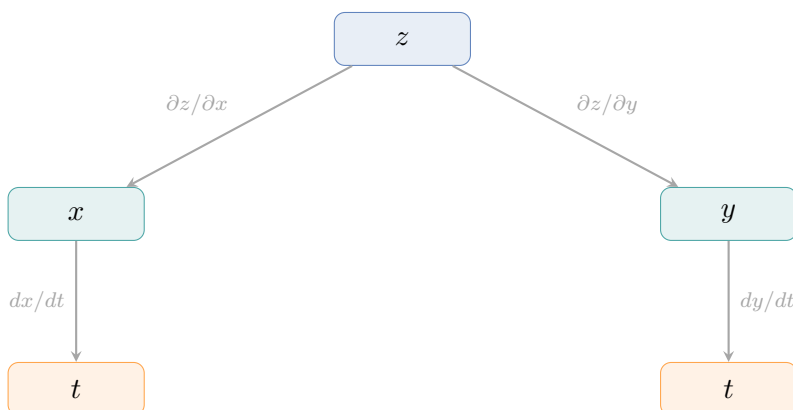
(b) Em  $(u, v) = (-2, 0)$ :

$$\frac{\partial w}{\partial u} = \frac{2}{-2} = \boxed{-1} \quad \frac{\partial w}{\partial v} = \boxed{2}$$

## Exercícios 13–15: Diagrama e fórmula da regra da cadeia

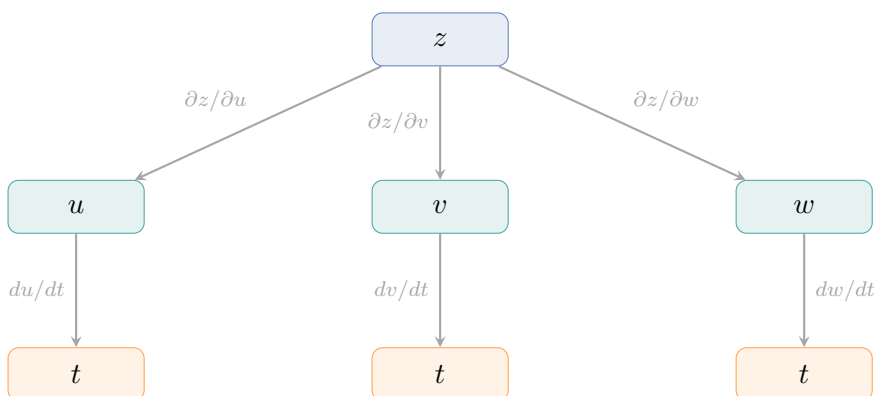
Nos diagramas abaixo: **azul** = função dependente, **verde** = variáveis intermediárias, **laranja** = variáveis independentes. As arestas indicam as derivadas parciais (ou ordinárias) em cada nível.

**Exercício 13**  $\frac{dz}{dt}$  para  $z = f(x, y)$ ,  $x = g(t)$ ,  $y = h(t)$



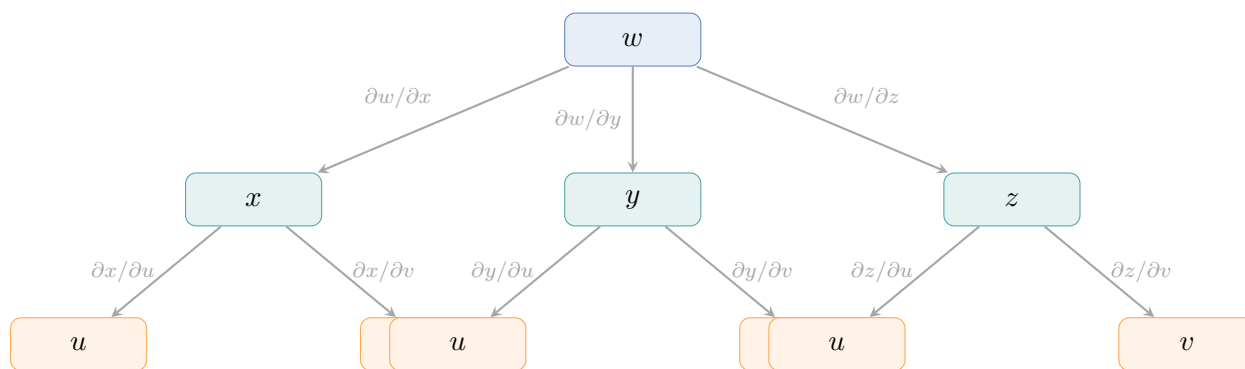
$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

**Exercício 14**  $\frac{dz}{dt}$  para  $z = f(u, v, w)$ ,  $u = g(t)$ ,  $v = h(t)$ ,  $w = k(t)$



$$\frac{dz}{dt} = \frac{\partial z}{\partial u} \frac{du}{dt} + \frac{\partial z}{\partial v} \frac{dv}{dt} + \frac{\partial z}{\partial w} \frac{dw}{dt}$$

**Exercício 15**  $\frac{\partial w}{\partial u}$  e  $\frac{\partial w}{\partial v}$  para  $w = h(x, y, z)$ ,  $x = f(u, v)$ ,  $y = g(u, v)$ ,  $z = k(u, v)$



$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial u}$$

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial v} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial v}$$